Simulations of Pseudomorphic GaAs/AlGaAs HEMTs with Extension into 2D-DCC Heterostructure Characterization

EE457 Course Project

Saniya Khinvasara (23B1268), Jigar Mehta (23B1273)
Department of Electrical Engineering, Indian Institute of Technology Bombay, India
Prof. Apurba Laha

***Abstract*—This study reports on the simulation and characterization of a pseudomorphic GaAs/AlGaAs HEMT heterostructure. Initially, TCAD Simulator was used to model the epitaxial layers, informed by a literature review for optimal structure design. A complete electrical characterization follows, analyzing band structure and performance via I-V characteristics under bias and doping variations.**

The work then extends to 2D-DDCC simulation for advanced analysis, requiring the configuration of the device geometry, contacts, mobility models, Schrödinger solver, and meshing. The primary goal is to compare the insights of the 2D model with the 1D results, demonstrating the necessary skills for both foundational and advanced heterostructure simulation and characterization.

***Index Terms*—HEMT, GaAs/AlGaAs, Pseudomorphic Heterostructure, Semiconductor Simulation, TCAD, 2D-DDCC, Electrical Characterization, I-V Characteristics, Band Structure**

I. Introduction

A. Motivation and Background

A high electron mobility transistor (HEMT) is a heterojunction device that exploits the high electron mobility and carrier drift velocity of the two-dimensional electron gas (2DEG) formed at a lattice-matched or strained heterointerface. Typically, this interface occurs between $Al_xGa_{1-x}As$ and undoped GaAs or $In_xGa_{1-x}As$.

Through modulation doping, extremely high carrier mobility can be achieved, which is valuable for device applications and the study of two-dimensional semiconductor effects.

High Electron Mobility Transistors (HEMTs) are essential devices in modern microwave and millimeter-wave electronic circuits, valued for their superior high-frequency performance and low-noise characteristics. This performance is achieved by utilizing a heterojunction between materials like GaAs and AlGaAs to form a Two-Dimensional Electron Gas (2DEG) at the interface, which facilitates high-speed electron transport. The specific structure, including epitaxial layers and doping, heavily influences the device characteristics.

Numerical device simulation is critical for efficient, cost-effective design, allowing for the rapid evaluation of structural changes and material properties.

B. Project Objectives

This project aims to conduct a comprehensive simulation study of a pseudomorphic GaAs/AlGaAs HEMT, progressing from foundational 1D modeling to advanced 2D characterization. The specific objectives are:

- To establish and characterize a basic GaAs/AlGaAs HEMT structure using the TCAD Simulator.
- To perform a complete electrical characterization of the structure, and performance evaluation through I-V characteristics under various bias and doping conditions.
- To extend the analysis to the 2D-DDCC Simulator to demonstrate the setup of complex device physics and geometry, specifically configuring the device profile, contacts, mobility models, Schrödinger solver, and meshing details.

C. Paper Organization

The remainder of this report is organized as follows: Section II reviews the relevant literature on HEMT design, Section III details the methodology for both the TCAD simulation setups, including material parameters and configuration steps. Section IV summarises the simulation results. Section V has the simulation results for non-pseudomorphic HEMTs and a comparison between the two. Finally, Section VI concludes the work and suggests avenues for future research.

II. Literature Review

Property	**GaAs**	**$Al_{0.3}Ga_{0.7}As$**	**$In_{0.3}Ga_{0.7}As$**
Bandgap E_g (eV)	1.424	1.90	0.95
Lattice constant a (Å)	5.65325	5.6501	5.785
Electron effective mass m_e^*/m_0	0.063	0.088	0.041
Hole effective mass m_h^*/m_0	0.51	0.585	0.45
Relative permittivity ε_r	12.9	12.05	13.9
Electron mobility μ_n ($cm^2V^{-1}s^{-1}$)	8500	3000	12000
Hole mobility μ_p ($cm^2V^{-1}s^{-1}$)	400	200	350
Electron affinity χ (eV)	4.07	3.74	4.55
Conduction band offset ΔE_c (eV)	—	0.24–0.28	∼0.20
Valence band offset ΔE_v (eV)	—	0.10–0.12	∼0.13

TABLE I
Material properties of GaAs, $Al_{0.3}Ga_{0.7}As$, and $In_{0.3}Ga_{0.7}As$ [1]

Pseudomorphic high electron mobility transistors (p-HEMTs) rely on strained InGaAs channels and modulation doping to achieve high electron mobility, strong carrier confinement, and excellent high-frequency performance. Early analytical and experimental work laid the foundation for

understanding the behavior of the two-dimensional electron gas (2DEG) formed within these heterostructures. Chang developed one of the first comprehensive charge-control and transport models for AlGaAs/GaAs and pseudomorphic AlGaAs/InGaAs HEMTs. His work systematically extracted mobility, access resistances, and saturation velocity from DC I–V characteristics, establishing methodologies still used for device characterization. Complementing these modeling approaches, Störmer provided a detailed physical description of the scattering mechanisms—remote ionized impurities, background impurities, interface roughness, alloy disorder, and phonons—that govern 2DEG mobility. These insights clarified the trade-offs in spacer thickness, doping profiles, and growth cleanliness, which remain central to p-HEMT optimization.[3]

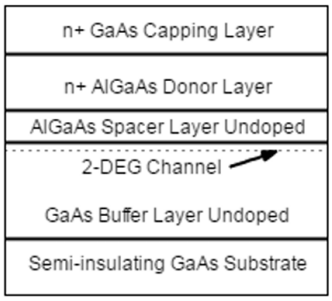


Fig. 1. Layer Structure of a conventional GaAs HEMT[2]

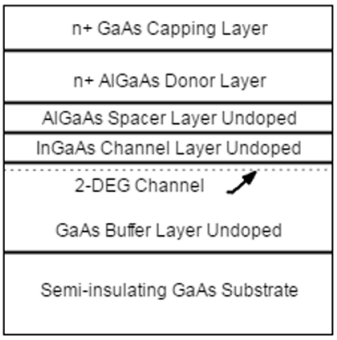


Fig. 2. Layer Structure of a pseudomorphic GaAs based HEMT[2]

The behavior of the 2DEG under lithographic patterning and heavy doping has also been extensively investigated. Allen, Stormer and Hwang (1983) demonstrated that small-area patterned regions of the 2DEG exhibit dimensional resonance effects and are highly sensitive to processing-induced changes in carrier density and scattering. Their results underscore that even minor lithographic steps can degrade mobility and affect high-frequency performance. For heavily doped or delta-doped channels, conventional transport measurements often fail; addressing this challenge, Mena et al. introduced a light-modulated Shubnikov–de Haas technique capable of extracting quantum lifetime and mobility in degenerate structures. Their findings show that although delta-doping enhances sheet charge density, it simultaneously increases impurity scattering and reduces quantum lifetime—highlighting design compromises in high-current p-HEMT architectures.[1]

Finally, modern simulation-based studies continue to refine understanding of p-HEMT behavior through numerical modeling. The InGaP/InGaAs/GaAs dual-channel HEMT study provided a TCAD-based assessment of gate-length scaling, current drive capability, and temperature dependence. The results demonstrated that shorter gate lengths reduce parasitic resistances and improve drain current, while elevated temperatures degrade mobility and device transconductance. Such simulation studies help validate material choices and heterostructure configurations prior to fabrication, improving the efficiency of device optimization.

Together, these works build a comprehensive understanding of p-HEMT operation—from fundamental band parameters and scattering physics to interface quality, processing effects, and numerical design tools. They collectively define the physical and technological considerations essential for optimizing pseudomorphic HEMTs for high-speed and low-noise applications.[4]

Double heterojunction pseudomorphic HEMTs (p-HEMTs) utilizing GaAs/AlGaAs/InGaAs structures offer a significant pathway for high-speed digital circuit applications by exploiting the enhanced electron transport properties of a strained InGaAs channel. By inserting a thin InGaAs layer between AlGaAs donor and spacer layers, these devices achieve improved carrier confinement and higher sheet charge density compared to conventional AlGaAs/GaAs heterojunctions due to larger conduction band discontinuities. Simulations of such a device with an 80 nm gate length characterize it as a depletion-mode transistor capable of achieving a cut-off frequency of 80 GHz. The suitability of this technology for digital logic has been further substantiated through the successful modeling and implementation of fundamental logic gates, including Inverter, NAND, and NOR circuits, confirming its potential for low-noise, high-frequency communication systems.[1][4][5]

III. Device Structure and Simulation Methodology

The pseudomorphic High Electron Mobility Transistor (pHEMT) was simulated using TCAD. The device consists of a multilayer heterostructure made of $Al_{0.22}Ga_{0.78}As$, $In_{0.18}Ga_{0.82}As$, GaAs, and an oxide layer. A non-uniform vertical mesh was used to accurately capture the strong band bending and the formation of the two-dimensional electron gas (2DEG) at the AlGaAs/InGaAs heterointerface.

A. Mesh Definition

The mesh was defined using the `mesh` command with extremely fine spacing near the active channel region. The vertical mesh spacing was reduced to as small as $0.0005\,\mu$m near $y = 0.035{-}0.038\,\mu$m, ensuring high resolution around the heterojunction where the 2DEG forms. Larger mesh spacing was used in regions far from electrical activity to reduce computational cost.

B. Material Regions

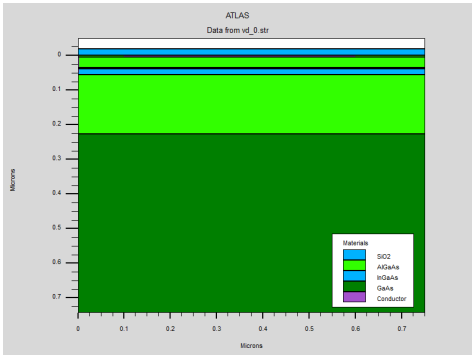


Fig. 3. Structure of the p-HEMT

The layer stack used in the simulation is summarized below:

- Oxide: $-0.02 < y < 0\,\mu$m
- $Al_{0.22}Ga_{0.78}As$ barrier layers: $0 < y < 0.038\,\mu$m
- $In_{0.18}Ga_{0.82}As$ channel: $0.038 < y < 0.056\,\mu$m
- $Al_{0.22}Ga_{0.78}As$ buffer: $0.056 < y < 0.226\,\mu$m
- GaAs semi-insulating substrate: $0.226 < y < 0.743\,\mu$m

These values define the conduction band offsets and drive 2DEG confinement in the InGaAs channel.

C. Doping Profile

A modulation-doped structure was implemented:

- Donor layer 1: $1 \times 10^{18}\,\text{cm}^{-3}$ in AlGaAs
- Donor layer 2 (delta-like): $5\times10^{18}\,\text{cm}^{-3}$ in AlGaAs (near 2DEG)
- InGaAs channel doping: $1 \times 10^{15}\,\text{cm}^{-3}$
- AlGaAs buffer doping: $1 \times 10^{16}\,\text{cm}^{-3}$
- Highly doped source/drain access regions: $1\times10^{18}\,\text{cm}^{-3}$

A fixed negative interface charge was applied at the oxide/AlGaAs interface: $Q_f = -1 \times 10^{12}\,\text{cm}^{-2}$

which enhances 2DEG formation and shifts the threshold voltage.

D. Physical Models and Numerical Method

The following ATLAS models were enabled:

- Shockley–Read–Hall recombination (SRH)
- Field-dependent mobility (FLDMOB)
- Concentration-dependent mobility (CONMOB)
- Bandgap narrowing (BGN)
- Fermi-Dirac carrier statistics (FERMI)

The system was solved using the Gummel–Newton iterative method with trap handling:

```
method gummel newton trap itlim=20
```

E. Bias Sweep Procedure

Two independent sweeps were performed:

a) 1. Output Characteristics (I_D-V_D):: $V_G = 0\,\text{V}, V_D : 0.05\,\text{V} \rightarrow 2.0\,\text{V}$

b) 2. Transfer Characteristics (I_D-V_G):: $V_D = 0.1\,\text{V}, V_G : -1.0\,\text{V} \rightarrow 1.2\,\text{V}$

IV. Simulation Results

A. Output Characteristics (I_D-V_D)

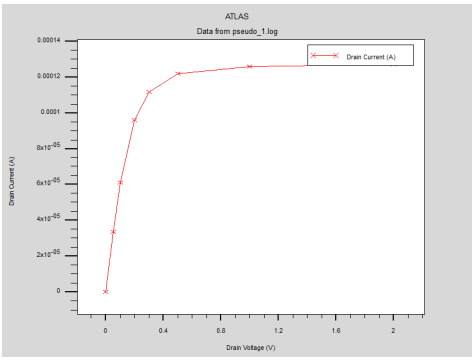


Fig. 4. Id-Vd plot

The linear I_D-V_D curve shows:

- A rapid increase in drain current in the linear region ($V_D < 0.2\,\text{V}$),
- A clear saturation region beginning near $V_D \approx 0.2\,\text{V}$,
- Peak saturation current of approximately $I_{D,\text{sat}} \approx 1.2 \times 10^{-4}\,\text{A}$

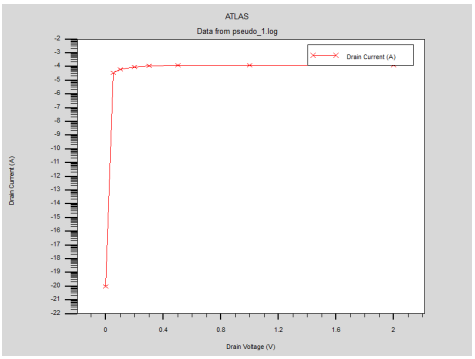


Fig. 5. log(Id)-Vd plot

The logarithmic plot confirms correct exponential behaviour at low V_D values and stable saturation.

B. Transfer Characteristics (I_D-V_G)

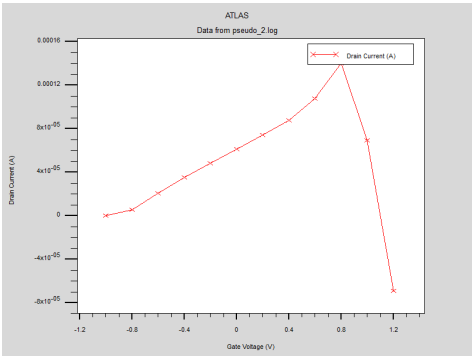


Fig. 6. Id-Vg plot

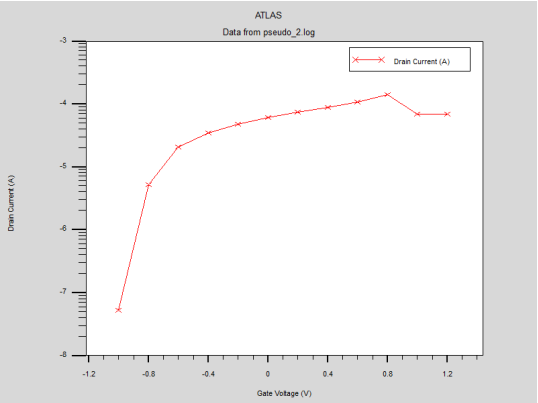


Fig. 7. log(Id)-Vg plot

At $V_D = 0.1\,\text{V}$, the transfer characteristics exhibit:

- Strong channel depletion for $V_G < -0.6\,\text{V}$,
- A steady increase in I_D up to $V_G \approx 0.8\,\text{V}$,
- A decrease beyond $V_G > 0.8\,\text{V}$, typically due to mobility degradation or drain-induced effects.

From the subthreshold region, the threshold voltage is approximately: $V_{\text{th}} \approx -0.5\,\text{V}$

C. g_m-V_G Characteristics

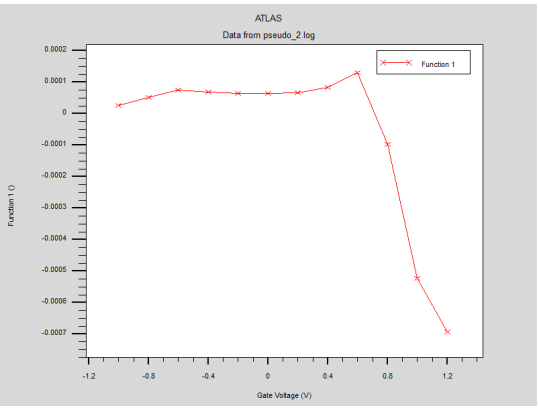


Fig. 8. g_m-V_G plot

The plot shows the transconductance g_m remaining nearly constant for negative V_G, peaking near $V_G \approx 0.5\,\text{V}$, and then rapidly decreasing as the gate enters forward-bias. This is typical behaviour of a pseudomorphic HEMT with a Schottky gate.

D. Device Structure Visualization

The plotted device cross-section clearly shows the oxide, the thin modulation-doped AlGaAs layers, the strained InGaAs channel layer, and the GaAs substrate. The thin AlGaAs donor layers facilitate the formation of the 2DEG at the heterointerface.

E. Summary of Findings

- A well-confined 2DEG channel forms at the AlGaAs/InGaAs interface.
- The device shows distinct linear, saturation, and cutoff regions characteristic of a functional pHEMT.
- Threshold voltage: $V_{\text{th}} \approx -0.5\,\text{V}$
- Saturation current: $I_{D,\text{sat}} \approx 1.2 \times 10^{-4}\,\text{A}$
- Both the I_D–V_D and I_D–V_G plots match expected pHEMT behaviour with modulation doping and heterostructure confinement.
- Transconductance peaked at $V_g \approx 0.6V$ with a value of $g_m \approx 0.00015$.

V. Some more Simulations

A. Conventional AlGaAs-GaAs HEMTs

Following are the results of the simulations done for conventional AlGaAs-GaAs HEMTs, for two different values of concentration of Al. The increase in the Al concentration in AlGaAs increases the strain between the two layers by Vegard's Law of linearity in lattice parameters. This serves as a transition to the pseudomorphic state of HEMTs.

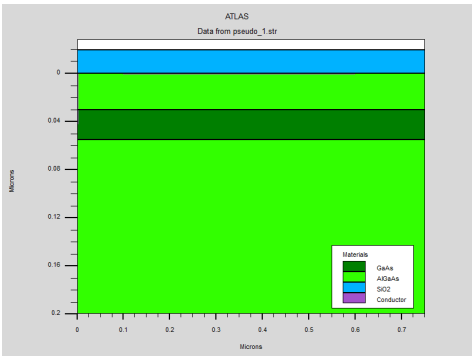


Fig. 9. Structure of the HEMT

1) GaAs-AlGaAs HEMT with x=0.3:

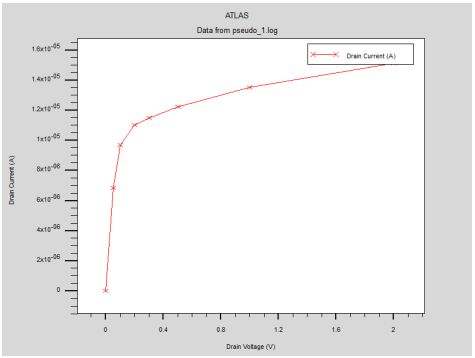


Fig. 10. Id-Vd plot

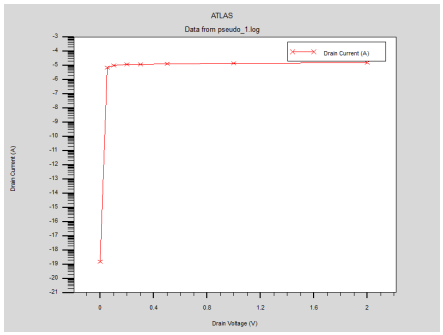


Fig. 11. log(Id)-Vd plot

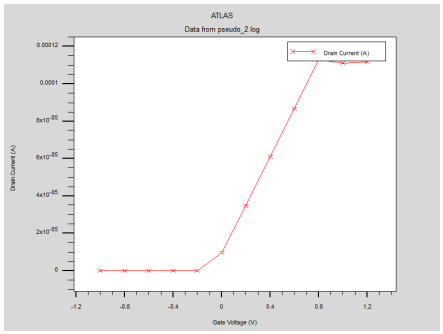


Fig. 12. Id-Vg plot

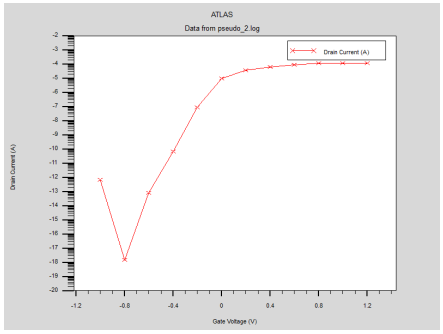


Fig. 13. log(Id)-Vg plot

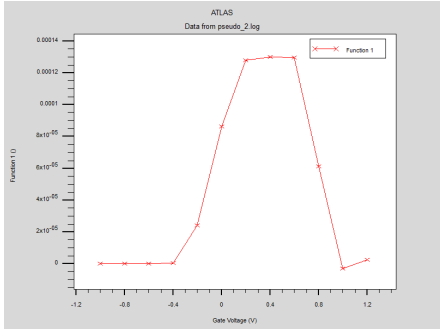


Fig. 14. gm-Vg plot

2) GaAs-AlGaAs HEMT with x=0.8:

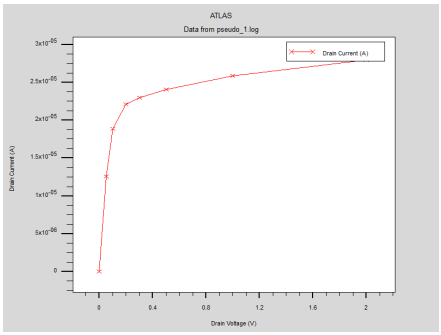


Fig. 15. Id-Vd plot

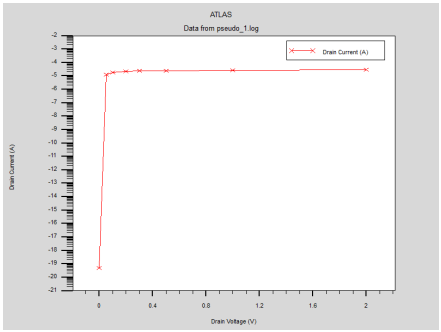


Fig. 16. log(Id)-Vd plot

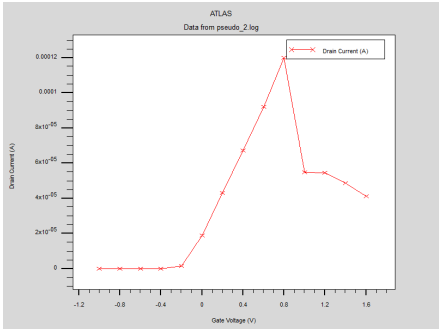


Fig. 17. Id-Vg plot

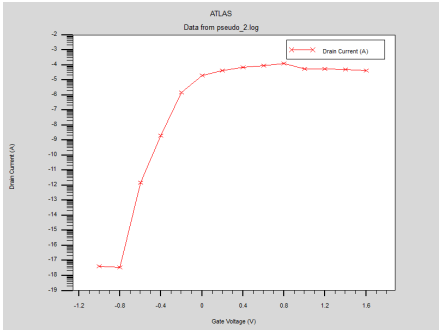


Fig. 18. log(Id)-Vg plot

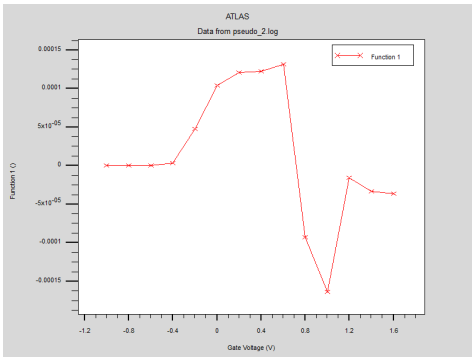


Fig. 19. gm-Vg plot

Transition from $Al_{0.3}GaAs$ HEMTs to $Al_{0.8}GaAs$ HEMTs and Finally to InGaAs pHEMTs

As device technology evolves from conventional $Al_{0.3}GaAs$/GaAs HEMTs to higher Al-content $Al_{0.8}GaAs$ HEMTs, and ultimately to strained-channel InGaAs pHEMTs, each material modification enhances the conduction-band offset, carrier confinement, and overall device performance till we can keep the strain below of what it will be at critical thickness.

1) Output Characteristics (I–V Behavior): $Al_{0.3}GaAs$ HEMTs exhibit moderate 2DEG density, resulting in limited drain current, higher knee voltage, and weaker saturation. Increasing the Al content to $Al_{0.8}GaAs$ deepens the quantum well and improves confinement, leading to slightly higher I_D and better saturation. The transition to an InGaAs pHEMT introduces a strained high-mobility channel, producing the highest $I_{D,\max}$, lowest knee voltage, and the flattest saturation region.
2) Transconductance (g_m): $Al_{0.3}GaAs$ devices provide modest g_m due to GaAs mobility limitations. $Al_{0.8}GaAs$ improves confinement and offers slightly higher g_m, but mobility is restricted. With the introduction of a strained InGaAs channel, electron mobility increases sharply, giving pHEMTs a significantly higher peak g_m at lower V_{GS}.
3) Threshold Voltage (V_{th}): $Al_{0.3}GaAs$ HEMTs have a relatively higher (less negative) threshold voltage. Increasing the Al fraction to 0.8 makes V_{th} (-0.5V) slightly more negative due to the enhanced band offset. InGaAs pHEMTs exhibit the most negative V_{th} because the strained channel forms a dense 2DEG with minimal gate bias.
4) RF Performance (f_T and $f_{\max}$): $Al_{0.3}GaAs$ HEMTs show limited RF performance due to mobility constraints and parasitic capacitances. Higher Al-content $Al_{0.8}GaAs$ improves these metrics slightly. In contrast, the strained InGaAs channel in pHEMTs yields dramatically higher mobility, lower gate capacitances, and superior confinement, resulting in the highest f_T and $f_{\max}$ among the three technologies.

Overall, increasing the Al concentration from 0.3 to 0.8 primarily enhances quantum-well depth and carrier confinement but remains constrained by GaAs mobility and trap-related effects. The final transition to a strained InGaAs channel represents a major leap in device physics, enabling much higher mobility, greater sheet charge density, improved RF metrics, and substantially better noise performance. This progression clearly demonstrates why modern high-frequency and high-performance III-V technologies predominantly favor pseudomorphic InGaAs HEMT structures.

B. Another potential p-HEMT structure

We tried to simulate the characteristics of a potential p-HEMT structure[3].

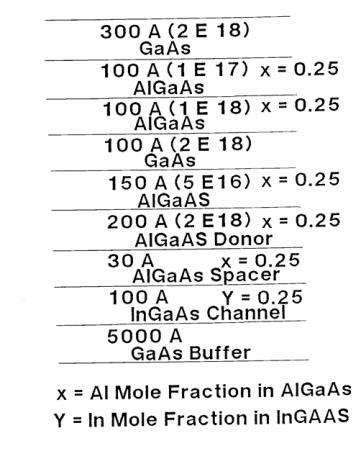


Fig. 20. Layer Structure of the p-HEMT [3]

- The structure consists of an $In_{0.25}Ga_{0.75}As$ channel sandwiched between undoped GaAs and modulation-doped $Al_{0.25}Ga_{0.75}As$ layers, forming a high-mobility 2DEG.
- A thin AlGaAs spacer layer separates the Si-donor-doped AlGaAs from the channel, reducing impurity scattering and improving mobility.
- The top GaAs cap provides surface passivation and forms the Schottky gate contact required for channel modulation.

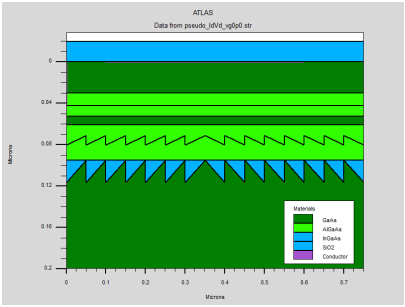


Fig. 21. Structure of the p-HEMT

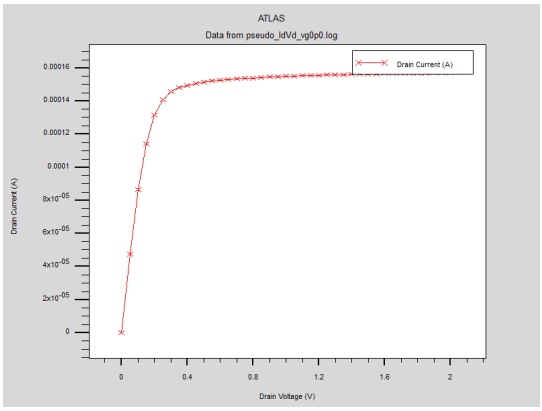


Fig. 22. Id-Vd plot

- The drain current increases rapidly at low V_D (linear region) and then saturates, demonstrating proper channel formation in the InGaAs quantum well.
- The early onset of saturation reflects high electron mobility and strong velocity saturation typical of pseudomorphic HEMTs.
- The smooth, monotonic behaviour confirms stable device operation with minimal short-channel effects for the simulated structure.

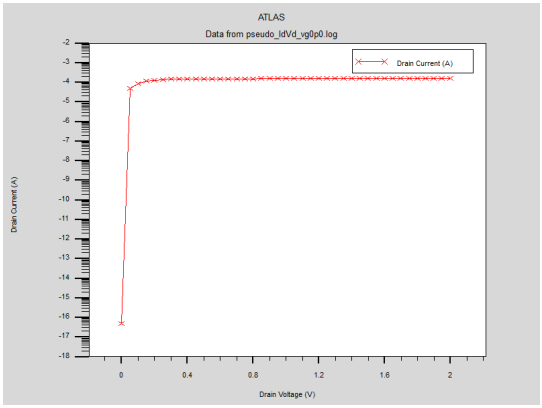


Fig. 23. log(Id)-Vd plot

- The logarithmic plot highlights the exponential rise of current at low V_D, corresponding to the linear-to-saturation transition.
- The near-constant $\log(I_D)$ at higher V_D confirms well-defined current saturation and stable channel transport.
- The steep slope at very low V_D originates from thermionic and field-assisted injection mechanisms dominant in this region.

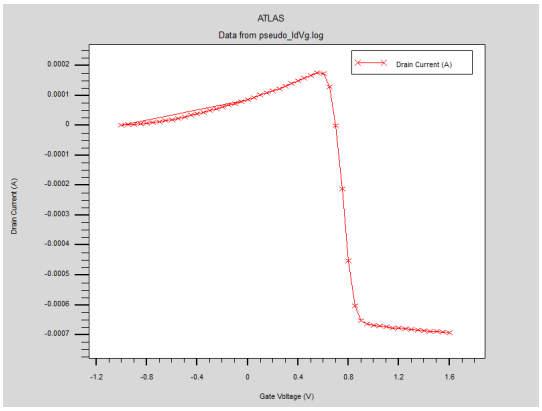


Fig. 24. Id-Vg plot

- The drain current increases with V_G due to enhanced 2DEG density in the InGaAs channel, reaching a maximum near the gate overdrive point.
- A sharp drop in I_D after the peak suggests depletion of the channel or onset of gate-induced barrier modulation.
- The overall curve is consistent with expected modulation of the 2DEG by the Schottky gate in a pHEMT structure.

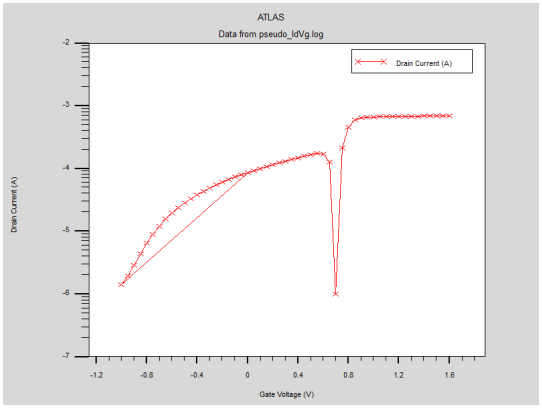


Fig. 25. log(Id)-Vg plot

- The logarithmic plot highlights the exponential rise of current at low V_D, corresponding to the linear-to-saturation transition.
- The near-constant $\log(I_D)$ at higher V_D confirms well-defined current saturation and stable channel transport.
- The steep slope at very low V_D originates from thermionic and field-assisted injection mechanisms dominant in this region.

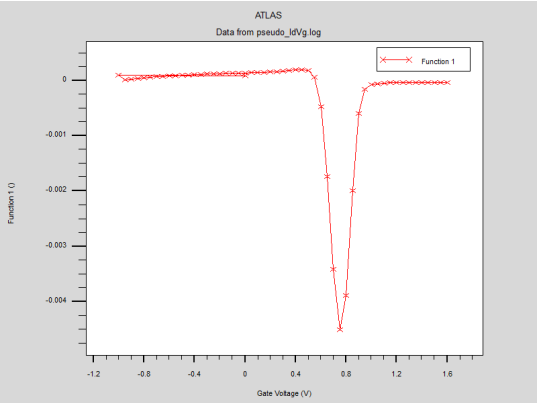


Fig. 26. gm-Vg plot

- The transconductance exhibits a sharp dip around $V_G \approx 0.7$–0.8 V, indicating a sudden change in the slope of the I_D–V_G curve due to strong field effects or trap-related behaviour at the AlGaAs/InGaAs interface.
- The small values of gm at lower and higher gate voltages correspond to the subthreshold and strongly accumulated regions, respectively, where the drain current becomes weakly sensitive to V_G.
- The negative sign of gm arises from the simulator's current sign convention (drain current plotted negative), and does not represent a physically negative transconductance.

VI. Conclusion

In this study, we explored the transition from a conventional AlGaAs/GaAs HEMT structure to pHEMT incorporating a strained InGaAs channel layer. Our experimental results clearly demonstrate the benefits of introducing strain into the channel. In particular, the pHEMT structure exhibited a noticeably higher drain current and a significantly enhanced transconductance g_m, both of which arise from the increased electron mobility and stronger carrier confinement provided by the strained InGaAs quantum well. These improvements directly translate into better switching capability, stronger channel control, and more efficient current modulation within the device.

A key takeaway from our transition analysis is that even though both device types share similar overall architectures, the introduction of a thin pseudomorphic InGaAs layer fundamentally alters the transport physics. The deeper conduction-band discontinuity and higher 2DEG density in the pHEMT enable faster electron motion and reduced channel resistance. This explains the performance enhancement seen in our measurements, highlighting how material engineering and strain engineering can substantially elevate device characteristics beyond conventional GaAs-based HEMTs.

While our work did not focus on full RF characterization, it is important to note that the same improvements—higher mobility, higher I_D, and higher g_m—are precisely the attributes that make pseudomorphic HEMTs valuable in high-speed and high-frequency applications. These characteristics also position pHEMTs as promising candidates for advanced digital and mixed-signal circuits, where fast switching and low-voltage operation are increasingly critical. As semiconductor scaling continues to challenge traditional technologies, strained-channel HEMTs such as the InGaAs-based pHEMT represent a strong pathway toward next-generation high-performance electronic devices.

In comparison to commercial TCAD packages, the 2D-DDCC software offers a more interactive set of visualization tools, allowing easier inspection of band diagrams, carrier distributions, and electrostatic profiles. Although TCAD generally provides faster simulation performance and more advanced physical models, 2D-DDCC excels in flexibility and user-controlled exploration of device parameters.

References

[1] I. Vurgaftman, JR Meyer, LR Ram-Mohan: *Band parameters for III–V compound semiconductors and their alloys*

[2] Parita Mehta, Lochan Jolly: *A GaAs/AlGaAs/InGaAs pseudomorphic HEMT structure for high speed digital circuits*

[3] Chujyh Chang: *Modelling and Characterisation of HEMT Devices*

[4] SJ Allen, Jr., HL Stormer, JCM Hwang: *Dimensional resonance of the two-dimensional electron gas in selectively doped GaAs/A1GaAs heterostructures*

[5] HL Stormer: *Electron mobilities in modulation-doped heterostructures*

[6] Giorgio Margaritondo: *Heterojunction band discontinuities: A fundamental problem in solid-state science*

[7] RA Mena, SE Schacham, EJ Haugland, SA Alterovitz, and PG Young, SB Bibyk and SA Ringel: *Characterization of the Transport Properties of Channel Delta-Doped Structures by Light-Modulated Shubnikov-de Haas Measurements*

[8] Limali Sahoo, Subhaluxmi Sahoo, Satyaprakash Narayan Das: *Simulation and Performance Evaluation of In Ga P/In Ga As/Ga As Dual Channel HEMT*